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(HFN)

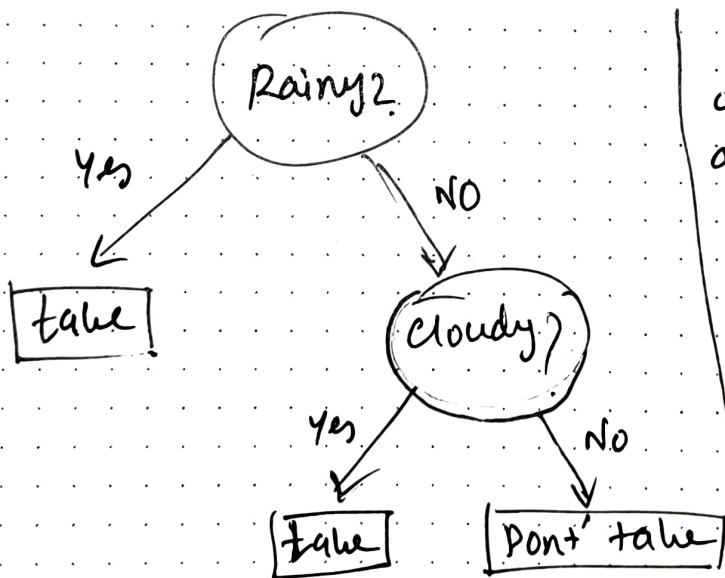
Decision Tree



① Refer to the booklet + slides first

Rainy	cloudy	Decision
Yes	Yes	Take
Yes	No	Take
No	Yes	Take
No	No	D. Take

Outcome → leaf Node



general
demonstration
of a dec. tree

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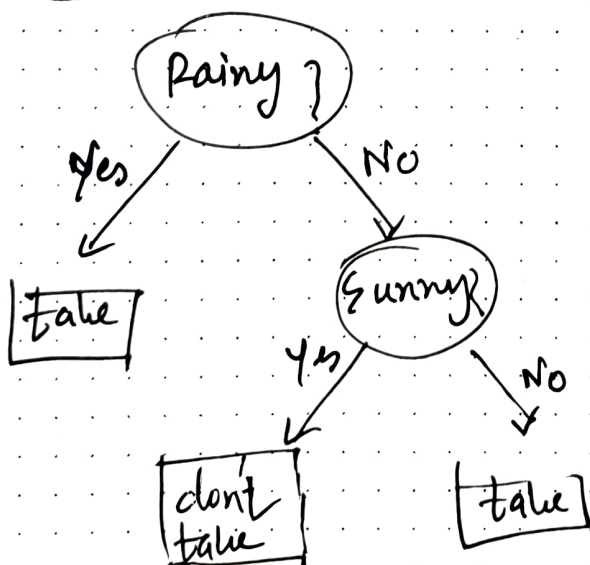


consider this example

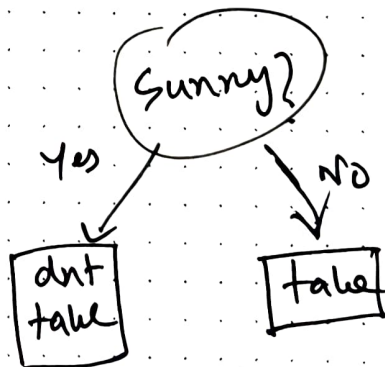
	Sunny	Rainy	Umbrella
Dataset 1	Yes	No	don't take
	No	Yes	take
	No	No	take

now, if we draw dec. tree with combination of features $\{x_1, x_2\} \Rightarrow y$ // $\{\text{Sunny, Rainy}\} \Rightarrow y$ in 2 variation $\Rightarrow$

Tree 1



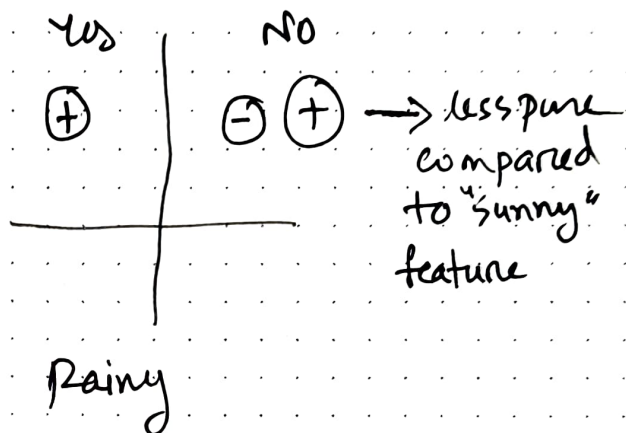
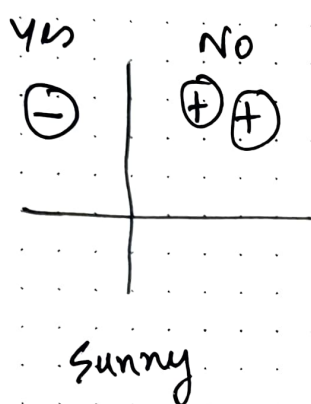
Tree 2



⊗ we don't need the other feature to conclude. So understanding which feature to be on the top level will help with the answer.

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Information Purity (Dataset 1)



* So, the purity of any feature must be calculated for efficient result. We use "Entropy" for it.

Entropy is the chaos/unstability within a feature.

high → less pure [→ impure for bin. classification]

low → more pure [→ pure " " "]

$$\# \text{ Entropy} = \sum_{i=1}^n -P_i \log_2(P_i)$$

i is the num. of class in this example 2 → take d.t. umbrella

→ this is two for our case since we always use binary classification for example solve

Entropy (sunny = yes)

$$\Rightarrow -P(D.T.) \log_2 P(D.T.) - P(T.U.) \log_2 P(T.U.)$$

[D.T. ⇒ Don't take umbrella,

T.U. ⇒ Take umbrella.]

✓

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$P(D.T.) \Rightarrow$ out of all the rows where
(sunny = yes), how many has the decision $\Rightarrow$ Don't
take umbrella

$$\Rightarrow P(D.T.) = \frac{1}{1} \quad \text{--- (1)}$$

$$\text{similarly, } P(T.U.) = \frac{0}{1} \quad \text{--- (11)}$$

plugging (1) & (2) $\Rightarrow$

$$-P(D.T.) \cdot \log_2 \left(\frac{1}{1} \right) - P(T.U.) \cdot \log_2 (P(T.U.))$$

$$= (-1/1) \log_2 (-1/1) - (0/1) \log_2 (0/1)$$

$= 0 \Rightarrow$ lowest entropy $\Rightarrow$ pure

similarly,

$\Rightarrow$ Entropy (sunny = low)

$$\Rightarrow P(D.T.) \cdot \log_2 (P(D.T.)) - P(T.U.) \cdot \log_2 (P(T.U.))$$

$$\Rightarrow 0/2 \log_2 (0/2) - 2/2 \log_2 (2/2)$$

$\Rightarrow 0 \Rightarrow$ lowest $\Rightarrow$ pure

If you calculate Entropy (rainy = no) = 1
" (rainy = yes) = 0

⊛ sunny is more pure since it reduces entropy to zero in every branch.

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Now, with this Entropy's concept, we'll discuss how to determine information gain.

Information gain of a feature x , given the decision/label is y is $\Rightarrow$

$$IG(x, y) = \text{Entropy}(y) - \text{weighted Entropy}(x) \quad \text{eqn 1}$$

(*) here, weighted entr. means the feature x may have two, three or more categories and each category may have different amount of samples. So we attach a weight with each category to make the affect for each category in the feature consistent. (*)

Note, this weight is nothing but the probability of that category among the whole feature. If the feature x has 3 category. (feature weather, cat $\rightarrow$ Rain, Sunny, Cold).

then weight of the category is $\Rightarrow \frac{x_i}{n}$ $\{x_i\}$

$$\text{So, } \left\{ \text{weighted Entropy}(x) = \sum_{i=1}^n p(x_i) * \text{Entropy}(x_i) \right\}$$

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Example (2) Feature

Sunny	Rainy	Decision
Yes	No	Dont
Yes	No	Dont
No	Yes	Take
No	No	Take
No	No	Take
No	Yes	Take

(eqn 1)

To know which feature gives more information gain

First calculate overall entropy of decision

Step 1

$$\text{Entropy}(y) = - \sum_{i=1}^n P_i \cdot \log_2(P_i)$$

$$\Rightarrow \left\{ -P_{\text{take}} \cdot \log_2(P_{\text{take}}) \right\} - \left\{ P_{\text{d.t.}} \cdot \log_2(P_{\text{d.t.}}) \right\}$$

$$\Rightarrow - \left\{ \left(\frac{4}{6} \right) \log_2 \left(\frac{4}{6} \right) + \left(\frac{2}{6} \right) \cdot \log_2 \left(\frac{2}{6} \right) \right\}$$

$$\Rightarrow 0.91$$

Now, what do understand from this?

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Entropy of the target/label/decision value is (0.91) in this case. Where

0 $\rightarrow$ pure

1 $\rightarrow$ impure

meaning, just from looking from the outside the distribution is more impure than pure. Not fully pure or fully mined. If the data was split as take vs don't take $\Rightarrow 3/3$; then it would've been fully mined (Entropy = 1); if the dataset had only one category present (all take/all don't take) then Entropy value would've been 0 [fully pure].

⊛ If fully pure, then no need for information gain, means the decision tree will just point to this 1 leaf node ~~as~~ with the value [take/don't take] with no added steps/level since all labels are same regardless

⊛ If fully mined, [impure] Then it's kind of a good sign for us means it's harder to separate the decision without splitting. So, we can add more feature (sunny or rainy) to help reduce the impurity in each branch now.

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Step 2

"Calculate weighted entropy split by sunny"

For

Sunny $\rightarrow$ yes: 2 sample, both "don't take" decision.

So class distribution is take: 0; d.t.: 2

Entropy (sunny = yes)

$$= -P(T) \log_2(P(T)) - P(D.T.) \log_2(P(D.T.))$$

$$\Rightarrow 0 - (2/2) \log_2(2/2)$$

$$\Rightarrow 1 \cdot \log_2 1 \Rightarrow 0$$

Similarly, Entropy of (Sunny = no)

for [sunny = no] $\rightarrow$ take: 4, dont take: 0

$En(\text{sunny} = \text{no}) = 0$ [same issue since no variation]

now,

$$\boxed{\text{Weighted Entropy (sunny)}} = \sum_{i=1}^n P(x_i) \cdot En(x_i)$$

$$= \{P(\text{sunny} = \text{yes}) \cdot En(\text{sunny} = \text{yes})\} + \{P(\text{sunny} = \text{no}) \cdot En(\text{sunny} = \text{no})\} \Rightarrow \underline{\underline{0}}$$

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Step 3

Information gain for feature "sunny"

$$I_4(\text{sunny}) = E_n(y) - \text{Weighted Entropy}(\text{sunny})$$
$$= 0.91 - 0 = \underline{\underline{0.91}}$$

Step 4

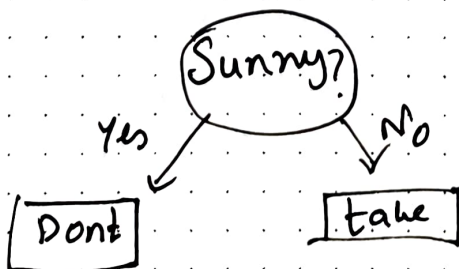
do the same step 2 again to

* Calculate the weighted entropy for Rainy.

$$I_4(\text{Rainy}) = 0.66$$

Step 5

now, comparing both $\Rightarrow I_4(\text{sunny}) > I_4(\text{Rainy})$
So, sunny is the better feature here. The tree will go like this: Choose this



⊛ Note: this row makes the decision split pure in each ~~node's~~ branch is a leaf node so completed.

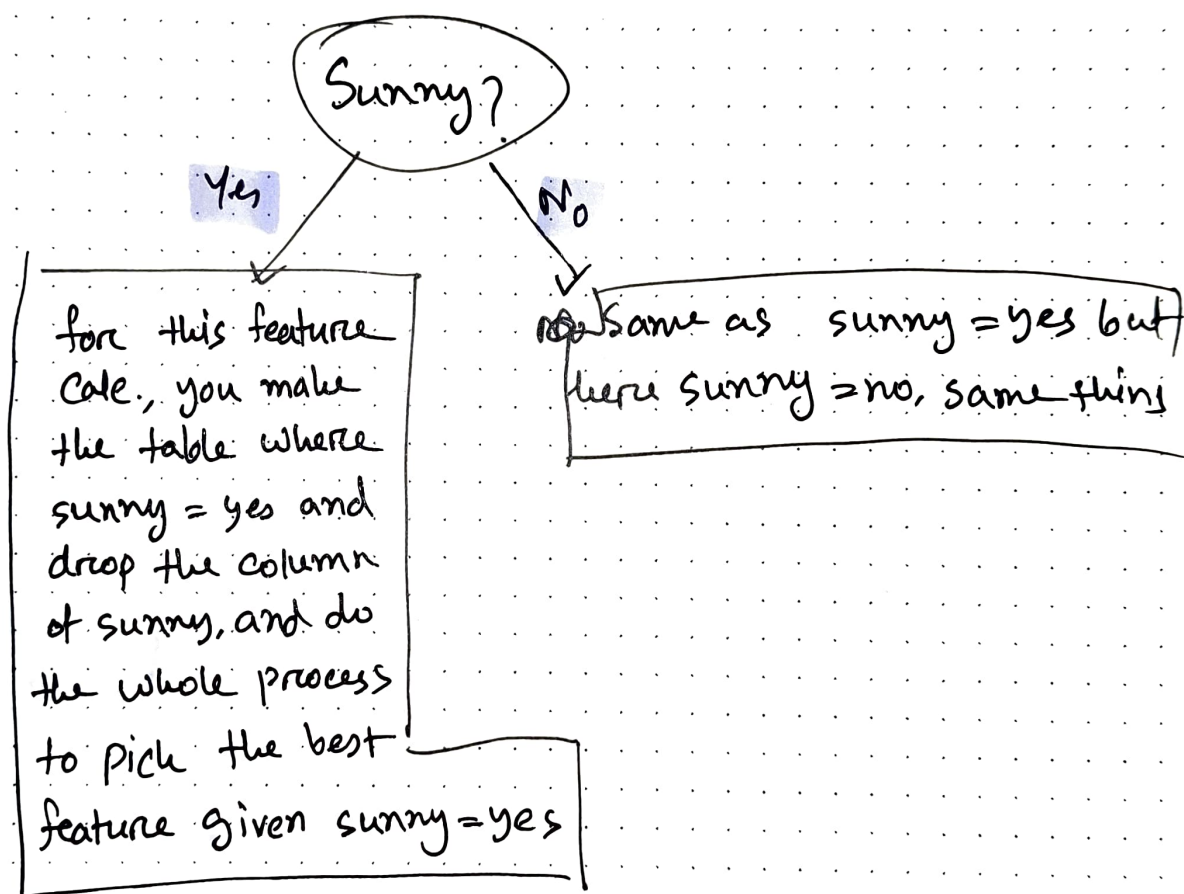
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Inspiring Excellence

But often in real life, the impurity will reduce slightly and you have to take the next best feature by isolating the table with the decision tree.

Example: (new example) Sunny, Rainy, storm etc. features.



⊗ When to stop this process

- node is pure/leaf (as the example)
- no feature left
- few examples remaining in the node.
- Max depth defined by you is reached
- Info gain is not incrementing much.